

EDUCATION

Master of Science in Applied Computing (MScAC) | University of Toronto

September 2026 – December 2027

Department of Computer Science, Faculty of Medicine • Focus in Artificial Intelligence in Healthcare

Courses: Neural Networks and Deep Learning, Computational Imaging and 3D Sensing, Automated Reasoning with Machine Learning

Honours Bachelor of Science (HBSc) | University of Toronto

September 2022 – June 2026

Department of Computer Science • Specialist in Bioinformatics & Computational Biology, major in Computer Science

GPA: 3.9/4.0 • Dean’s List Scholar (2022 – 2026) • B. Arthur Bensley Scholarship in Life Sciences (2025)

Courses: Machine Learning, Artificial Intelligence, Systems Biology, Data Structures and Algorithms, Probability and Statistics

RESEARCH EXPERIENCE

Computational Pathology Machine Learning Researcher | Princess Margaret Cancer Centre, UHN

September 2025 – Present

Computational Cancer Genomics Lab, Dr. Sushant Kumar: Automated Segmentation of IHC and H&E Staining

- Designed a multi-modal cascaded diffusion model integrating RNA, CNA, RPPA, and methylation foundation models to synthesize tumour histology from pan-cancer data, achieving 83% match in morphological diversity across cancer types using PyTorch Lightning
- Led development of a 24.4M-parameter U-Net segmentation pipeline and QuPath extensions trained on 10M patches for automated vessel segmentation and fibroblast quantification, achieving 92% Dice score and pathologist-approved CHIP risk-group classification
- Built and deployed a Python package for TCGA-native DNA methylation data engineering, enabling ML-ready preprocessing at HPC scale 26% faster than current benchmarked research toolkits, with automated reproducibility audits and logging using Python

Spatial Proteomics Research Intern | Lunenfeld-Tanenbaum Research Institute, Mount Sinai

May 2025 – February 2026

Campbell Lab, Dr. Kieran Campbell: Computer Vision Methods for Highly Multiplexed Biological Images

- Built the first end-to-end spatial proteomics toolkit (ResNet-34 encoder convolutional autoencoder) optimized for highly multiplexed IMC data, reducing runtime by 30% over existing non-optimized toolkits using Python and PyTorch Lightning; cited in paper at *Nature*
- Designed novel stratified, morphology-specific virtual ROI sampling protocol benchmarking TMA-IMC against WS-IMC as ground-truth for capturing tumour microenvironment spatial heterogeneity in PDAC using Python and PyTorch Lightning; cited in paper at *Nature*

Computational Biology Research Intern | Massachusetts General Hospital, Harvard Medical School

September 2024 – July 2025

Sadreyev Lab, Dr. Ruslan I Sadreyev: Molecular Mechanisms of Acute Myeloid Leukemia

- Engineered multi-modal AML sequencing workflows integrating the largest cross-project methylation/RNA-seq cohort to date (TCGA, TARGET, BEATAML1/2), applying fuzzy K-means clustering ($K=3$ DMGs, $K=5$ DEGs) for integrative differential subtype analysis using R
- Identified three novel candidate biomarkers and potential drivers of aggression in the largest AML molecular subtype (NPM1, 30%)

PROFESSIONAL EXPERIENCE

Lead Systems Software Engineer | Victoria University, University of Toronto

May 2024 – Present

- Led end-to-end design, development, and deployment of a university-scale intranet platform, encompassing architectural redesign, full database restructure and migration, data retention protocol design, and Microsoft 365 migration using PostgreSQL and MongoDB
- Redesigned backend systems architecture and normalized database schema, reducing storage redundancy by 17% using PostgreSQL

SELECTED RESEARCH OUTPUTS

- αSMA-Fibroscore** (Python) – Tuned U-Net for vessel segmentation, αSMA+ fibroblast quantification, and CHIP risk stratification
- WSI-IMC-Workflows** (Python) – Toolkit for tissue mask generation, patch extraction, and preprocessing of whole-slide IMC data
- MethylTrain** (Python) – Python package for DNA methylation data engineering for research-grade machine learning workflows
- BioTokenizerR** (R) – R package for biology-aware BPE tokenization of RNA, DNA, and amino acid sequences optimized for big data

PUBLICATIONS

- Nowlan F, et al. (incl. Li SM), “Integrated spatial proteomics of human PDAC uncover an expanded tumour-immune-stroma spectrum with genomic associations,” *bioRxiv*, 2025 – under second-round peer-review at *Nature*
- Illamperuma AT, Li SM, Ghumman J, Gandhi B, "Validation and algorithmic fairness in AI-enabled ECG models for myocardial ischemia: A systematic review of North American evidence," *Journal of Biomedical Informatics*, 2026 – under first-round peer-review
- Shakfa N, et al. (incl. Li SM), “Single cell imaging uncovers a coordinated tumor-immune-stroma spectrum with genomic associations in PDAC,” *Cancer Research: AACR Annual Meeting*, 2026

TECHNICAL SKILLS

Languages: Fluent in Python, R, C/C++, Java, and PostgreSQL • Experienced in MySQL, TypeScript, JavaScript, Bash, Assembly

ML/DL: Fluent in PyTorch Lightning, TensorFlow, scikit-learn • Experienced with Machine Learning, Computer Vision, and Deep Learning